

CAR-engineering of innate and innate-like immune cells: a new horizon in adoptive cell therapy for solid tumors

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ABSTRACT

Background Chimeric antigen receptor (CAR) therapies have revolutionized cancer immunotherapy, particularly in hematologic malignancies, but their efficacy in solid tumors remains limited. Key barriers include tumor antigen heterogeneity, on-target/off-tumor toxicity, impaired trafficking, and an immunosuppressive tumor microenvironment.

Methods We conducted a narrative review of preclinical and clinical studies investigating CAR-engineered innate and innate-like immune cells, including CAR-natural killer, CAR- $\gamma\delta$ T, CAR-natural killer T (NKT), and CAR-macrophages, focusing on their biological features, therapeutic potential, and current clinical development in solid tumors.

Results These alternative platforms exhibit distinct advantages over conventional CAR-T cells, including reduced risk of severe toxicities, improved trafficking, overcoming antigen loss, and higher allogeneic potential. Emerging clinical data suggest favorable safety profiles, although limited persistence and variable efficacy remain key challenges. Advances in cell engineering, such as cytokine armoring and non-viral gene transfer, are further enhancing their therapeutic potential.

Conclusions CAR-engineered innate and innate-like immune cells represent a promising next-generation strategy to overcome the limitations of conventional CAR-T therapies in solid tumors. Among these, CAR-NKT and CAR- $\gamma\delta$ T cells may offer particular advantages for clinical translation, warranting further investigation in future trials.

INTRODUCTION

Conventional chimeric antigen receptor (CAR)-T therapies have transformed the treatment of hematologic malignancies, but they have shown limited success in solid tumors due to a combination of biological and clinical barriers.¹ Antigen selection is challenging because most targets are tumor-associated antigens (TAAs) with heterogeneous expression and potential on-target/off-tumor toxicity, enabling antigen escape and limiting safe dosing. In addition, abnormal vasculature and dense extracellular matrix restrict CAR-T cell trafficking, and the

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Adoptive cell therapies, particularly chimeric antigen receptor (CAR)-T cells, have demonstrated remarkable efficacy in hematologic malignancies, but have shown limited success in solid tumors due to biological and clinical barriers.

WHAT THIS STUDY ADDS

⇒ This review provides a comprehensive and updated overview of CAR-engineered innate and innate-like immune cells, highlighting their potential to overcome key limitations of conventional CAR-T cells in solid tumors.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE, OR POLICY

⇒ These findings support the development of next-generation CAR-based strategies and may guide future clinical trial design and translational research efforts in solid tumor immunotherapy.

tumor microenvironment (TME) promotes functional exhaustion and limits persistence. These biological constraints are compounded by clinical factors. Solid tumor patients are often older and frailer, with reduced bone marrow reserve, making lymphodepleting conditioning less tolerable. Moreover, rapid disease progression and manufacturing delays can hinder treatment administration.

To address these challenges, alternative CAR-based therapies are being developed, including CAR-natural killer (NK), CAR- $\gamma\delta$ T, and CAR-macrophages (CAR-M) cell therapies. Each of these cell types offers unique advantages in cancer immunotherapy. NK cells are highly adaptable and rapid-acting, making them ideal for targeting diverse cancers, especially those that evade adaptive immune responses. Their low side-effect profile further enhances their therapeutic potential. $\gamma\delta$ T cells can function in both innate and adaptive immune roles, providing a dual mode of action; they also demonstrate

resistance to immune exhaustion, increasing their efficacy in the TME. Macrophages, integral to shaping the TME, contribute through their capacity for tumor cell phagocytosis and immune activation.

This review aims at presenting a comprehensive state of the art of CAR-based therapies, discussing their benefits, limitations, and latest clinical advancements.

CAR-T CELLS

In recent decades, CAR-T cells have shown promising outcomes against hematologic malignancies. These advancements support CAR-T cell applications in solid tumors, prompting an increase in preclinical and clinical CAR-T cell trials targeting solid tumors.^{2,3}

Advantages and drawbacks

Genetic modification of polyclonal T cell populations enables the generation of TAAs-specific T cells within 2 weeks. This represents an advantage over the in vitro process of selecting, characterizing, and expanding T-cell clones with native specificity for major histocompatibility complex (MHC)-tumor peptide complexes.⁴ Furthermore, CARs facilitate T-cell recognition of antigens in an MHC-independent manner, effectively bypassing major immune escape mechanisms in tumors, such as MHC molecule downregulation, and are more adaptable in targeting surface structures on tumor cells, such as glycosylated proteins, gangliosides, and carbohydrate antigens.

Recent advances have enhanced CAR immune cells with additional features to improve efficacy and reduce toxicity. For instance, chemokine receptors can be added to promote T cell migration and infiltration into the TME, while pro-inflammatory cytokines can help overcome immunosuppression within the TME. Additionally, knocking out inhibitory receptors can prevent CAR-T cell exhaustion, and the inclusion of suicide or inhibitory genes can reduce toxicity.

A critical challenge in solid tumors is antigen selection. Ideal targets are tumor specific antigens (TSAs), exclusively expressed on malignant cells; however, most of the known antigens are TAAs, and off-tumor damage is common. Another barrier is antigenic heterogeneity: CAR-T cells eliminate tumor cells with high levels of the target antigen but may not eliminate cells with lower antigen expression. CAR-T cell trafficking and infiltration into tumor sites also pose challenges, largely due to the abnormal vasculature, the dense extracellular matrix, and the immunosuppressive TME. Tumor-associated macrophages, regulatory T (Treg) cells, myeloid-derived suppressor cells (MDSCs), and cancer-associated fibroblasts (CAFs) not only directly suppress CAR-T cell function but also produce immunosuppressive cytokines, creating a metabolically unfavorable microenvironment.⁵ CAR-T cell exhaustion, marked by a progressive loss of effector function, leads to a reduced proliferative capacity and diminished cytokine production, ultimately impairing the ability of CAR-T cells to eliminate tumor

cells.⁶ While CAR-T cell therapy has a high response rate, it also carries a considerable risk of side effects, typically cytokine release syndrome (CRS) and immune effector cell-associated neurotoxicity syndrome (ICANS). In addition to efficacy and side effect concerns, the preparation time and cost present challenges. Moreover, since leukocytes are often harvested after patients have undergone multiple treatments, the quality of their T cells may be compromised.

Advancements in solid tumors

Several clinical trials of CAR-T cells targeting solid tumors have been approved in recent years, with approximately 200 phase I/II trials currently registered (table 1).⁷ Regarding adverse events, patients experienced both localized and general side effects, including fever, fatigue, nausea, vomiting, and lung toxicity. The most common toxicities were CRS and ICANS, ranging from transient cognitive impairments to more severe effects (encephalopathy and seizures). A phase I/II clinical trial conducted at IRCCS Ospedale Pediatrico Bambino Gesù in Rome involving 27 patients with high-risk neuroblastoma treated with third-generation CAR-T cells reported a high response rate at 18 months post-infusion. Most patients experienced mild CRS, while all patients developed severe hematological toxicity, due to lymphodepletion, and hepatic toxicity.⁸

While clinical trials showed encouraging results, preclinical research is focused on innovative approaches for solid tumors treatment. For instance, allogeneic CAR-T cells have already been considered, and efforts are trying to reduce MHC disparity through gene editing technologies and alternative cell populations in order to avoid graft-versus-host disease (GvHD) and toxicities related to immune rejection.^{9,10} Currently, thirty clinical trials are utilizing allogeneic CAR-T cells for hematologic malignancies, and three are ongoing in solid tumors with allogeneic anti-MUC1 CAR-T against advanced solid tumors (NCT05239143) and allogeneic NKG2D-based CAR-T cells against metastatic colorectal cancers (NCT03692429 and NCT04991948). Primary results demonstrate that they are safe and well tolerated. Another valuable strategy could be targeting TME components and/or armored CARs.¹¹ Masking immunotherapy is another active field of research and includes all the strategies adopted to limit on-target off-tumor toxicities: examples are switchable and logic-gated CAR-T cells.¹² Finally, 'fast' CAR-T cells have been recently developed to shorten the length of drug's production from 2 to 4 weeks to approximately 1 week; this method consists in gathering CAR-T cells the day after transduction without an expansion step, thereby increasing the amount of T cells with a less differentiated and a more effector phenotype.¹³ Results of a clinical trial (NCT03825718) of F-anti-CD19 CAR-T cells in patients with B-cell acute lymphoblastic leukemia documented a younger cellular phenotype with a less exhausted profile and more effective in killing cancer cells; the 25 patients

Table 1 CAR-engineered cells against solid tumors, clinical trials with published results

ID	Antigen targeted	Solid tumor type	Phase	N° patients	Median FU (mo)	Antitumor efficacy	Safety/toxicity
CAR-T							
NCT03373097	GD2	Neuroblastoma	1/2	27	36	ORR 63%; mOS 60%, EFS 36%	CRS G1 94% patients
NCT04196413	GD2	Diffuse midline glioma	1	13	33	mOS 20.6 mo	7 CRS G2, 5 ICANS G2, TIAN G1 all patients
ACTRN12613000198729	GD2	Advanced solid tumor	1	14	–	5 PR, 2 SD, 5 PD, 2 non-evaluable	One CRS G1
NCT01869166	EGFR	Pancreatic cancer	1	16	–	mPFS 3 mo, mOS 4.9 mo; 4 PR, 8 SD	CRS and ICANS G3 all patients
NCT03182816	EGFR	Advanced solid tumors	1	9	–	1 PR, 6 SD, 3 PD; mPFS 7.13 mo, mOS 15.63 mo	CRS G1 to G3 all patients
NCT01935843	HER2	Biliary tract and pancreatic cancers	1	11	–	1 PR, 5 SD, 5 PD; mPFS 4.8 mo	10 CRS G1–2, one CRS G3
NCT00902044	HER2	Advanced sarcomas	1	13	8.2	3 CR and 4 SD; mPFS 4.2 mo, mOS 8.2 mo	9 CRS G1–2, 2 CRS G3–4
NCT02932956/ NCT04715191/ NCT05103631	GPC3	Pediatric solid tumors	1	18	–	4 PR, 7 SD, 7 PD; 33% ORR (4/12) for patients of the IL15.CAR-T group, no objective responses in the CAR-T group	CRS G1 one patient CAR-T group, CRS G1–3, 9 patients IL15.CAR-T group
NCT03873805	PSCA	Prostate cancer	1	14	–	7 SD and 7 PD	5 CRS G1–2
NCT02744287	PSCA	Prostate and pancreatic cancer	1	33	–	2 PR, 16 SD, 8 PD, and 9 non-evaluable	2 CRS G3–4, one ICANS G3, one fatal sepsis
NCT01583686	Mesothelin	Advanced solid tumors	1	15	–	1 SD and 14 PD	Other types of toxicities reported, no CRS or ICANS
NCT02414269	Mesothelin	Advanced primary and secondary pleural malignant diseases	1	27	20.3	2 PR, 9 SD and 5 PD in the combination group, no data on radiological evaluation for the CAR-T group; mOS 17.7 mo in the CAR-T group and 23.9 mo in the combination therapy group	No CRS or ICANS >G2
NCT01355965	Mesothelin	Pancreatic cancer	1	2 (case reports)	–	2 PR	No CRS or ICANS
NCT03198546	GPC3 and mesothelin	Advanced solid tumors	1	6	–	1 CR, 1 PR, 2 SD and 2 PD	No CRS >1

Continued

Table 1 Continued

ID	Antigen targeted	Solid tumor type	Phase	N° patients	Median FU (mo)	Antitumor efficacy	Safety/toxicity
NCT03874897	CLDN18.2	Advanced gastro-enteric solid tumors	1	98	32.4	1 CR, 37 PR, 53 SD, 7 PD; ORR 38.8%, DCR 91.8%; mPFS 4.4 mo, mOS 8.8 mo	CRS G1–2 96.9% patients;
NCT023449724	CEA	CRC	1	10	–	7 SD and 3 PD	No severe adverse events;
NCT02850536	CEA	Liver metastasis from pancreatic cancer	1	1	–	PR; mOS 23.2 mo	CRS G1–2 all patients
NCT03089203	PSMA	Prostate cancer	1	13	–	Best overall response SD, mPFS 4.4 mo, and mOS 15.9 mo	5 CRS G2, CRS G4 with death for one patient
NCT02706392	ROR1	Advanced hematological and solid tumors	1	21 (18 with solid tumors)	–	Two of the threepatients with CLL and one patient with TNBC achieved PR, 16 patients with TNBC or NSCLC achieved SD, 1 patient PD, and 1 patient dead before radiological assessment	11 CRS G1–2, 5 CRS G3, 2 ICANS G1
NCT06043466	CEACAM5	Advanced gastroenteric malignancies	1	14	–	7 SD and 7 PD	Other types of toxicities reported, no CRS or ICANS;
NCT02541370	CD133	HCC	2	21	–	1 PR, 14 SD, 6 PD; mOS 12 mo, mPFS 6.8 mo	Other types of toxicities reported, no CRS or ICANS;
NCT04438083	CD70	RCC	1	16	–	1 CR, 12 SD and 3 PD; mPFS 2.9 mo, mOS 20.5 mo; ORR 6.3%, DCR 81.3%	8 CRS G1–2, 4 CRS G3–4, no ICANS or GvHD;
NCT03060356	c-MET	Melanoma and TNBC	1	7	–	4 SD and 3 PD	One CRS G1
ChiCTR2000040645	Guanylate cyclase-C (GCC19)	CRC	1	15	–	ORR 40%; 6 PR, 5 SD, 4 PD; mDOR 5.1 mo, mPFS 6 mo, mOS 22.8 mo	CRS G1–2 in all patients, one ICANS G4
CAR-NKT							
NCT03294954	GD2	Neuroblastoma	1	12	–	2 PR and 1 SD; ORR 25% (3/12)	One CRS G2
CAR-NK							

Continued

Table 1 Continued

ID	Antigen targeted	Solid tumor type	Phase	N° patients	Median FU (mo)	Antitumor efficacy	Safety/toxicity
NCT03415100	NKG2DL	Colorectal cancer	1	3	–	All achieved SD, (one patient with CMR at liver metastasis in PET)	CRS G1 all patients
NCT03656705	PDL1	NSCLC	2	1 (case report)		SD	CRS G4
NCT04050709	PDL1	Advanced solid tumors	1	9	–	4 SD (Additionally, a CR was reported on a compassionate expanded use in a patient with metastatic pancreatic cancer)	No toxicities
NCT04390399	PDL1	Pancreatic cancer	2	63	–	mOS 6 mo	No toxicities
NCT03941457	ROBO1	Pancreatic cancer	1	1 (case report)	–	SD	CRS G1
NCT03383978	HER2	Glioblastoma	1	9	–	5 SD, 4 PD; mPFS 7 weeks, mOS 31 weeks	No toxicities
NCT03692663	PSMA	Prostate cancer	1	1 (case report)	–	SD, with PSA reduction of 97.5%	No toxicities
NCT02839954	MUC1	Advanced solid tumors	1	13	–	9 SD, 4 PD	No toxicities
CAR-M							
NCT06562647	Mesothelin	Ovarian cancer	1	2	1	SD	No toxicities
NCT04660929	HER2	Advanced solid tumors	1	14	2	4 SD; mPFS 1.47 mo	All CRS G1–2

CLDN, claudin; CLL, chronic lymphoblastic leukemia; c-MET, proto-oncogene mesenchymal-epithelial transition; CMR, complete metabolic response; CR, complete response; CRC, colorectal cancer; DCR, disease control rate; EFS, event-free survival; EGFR, epidermal growth factor receptor; FU, follow up; GD, ganglioside; GPC, glypican; HCC, hepatocellular carcinoma; HER2, human epidermal growth factor receptor 2; IL, interleukin; mDOR, median duration of response; mo, months; mOS, median overall survival; mPFS, median progression-free survival; MUC, mucin; NSCLC, non-small cell lung cancer; ORR, objective response rate; PD, progression disease; PDL, programmed death ligand; PET, positron emission tomography; PR, partial response; PSA, prostate specific antigen; PSCA, prostate stem cell antigen; RCC, renal cell carcinoma; ROBO, roundabout; ROR, receptor orphan receptor; SD, stable disease; TIAN, tumor inflammation-associated neurotoxicity; TNBC, triple negative breast cancer.

treated in this trial achieved excellent objective responses, and the toxicities were manageable.¹⁴

CAR-NK CELLS

CAR-NK cells represent an emerging frontier in adoptive cell therapies (ACTs). Although clinical evidence is still limited, this approach appears to offer several advantages over CAR-T cells in treating solid tumors.

NK cells are highly sensitive to the presence or absence of MHC-I molecules, which are abundantly expressed in normal cells to present self-antigens to T cells. They use inhibitory receptors, such as the killer-cell immunoglobulin-like receptor (KIR), to interact with MHC-I molecules. When a NK inhibitory receptor recognizes an MHC-I molecule, the NK cell is inhibited, blocking its cytotoxic response and sparing the target cell. However, in many viral infections or tumor transformations, abnormal cells downregulate MHC-I expression to evade immune surveillance. NK cells detect this downregulation as ‘non-self’ and kill these abnormal cells.

Additionally, they can recognize tumor neoantigens and various stress-induced proteins, such as MHC class

I chain-related molecules (MIC)-A, MIC-B, and ULBP (UL16-binding protein), through activating receptors including NKp46, NKp44, NKp30, NKG2D, and DNAM-1 (figure 1). These receptors trigger degranulation, releasing cytotoxic molecules like perforin and granzymes. Other pathways contributing to NK-mediated cytotoxicity include death receptor-mediated killing (through FasL and TRAIL), inhibition of tumor angiogenesis (releasing effector molecules such as interferon- γ (IFN- γ)), and antibody-dependent cell-mediated cytotoxicity (ADCC) via CD16.

Advantages and drawbacks

One significant advantage of CAR-NK cells is their lower toxicity. Some studies have shown that they induce less CRS, likely due to the different cytokines produced, including IFN- γ , granulocyte-macrophage colony-stimulating factor, and various chemokines such as C-C motif chemokine ligand 1 (CCL1), monocyte chemoattractant protein (MCP)-1, macrophage inflammatory protein (MIP)-1 α/β (CCL3/4), and CXCL8. However, due to the limited number of completed clinical studies,¹⁵ definitive conclusions are difficult to draw. For instance,

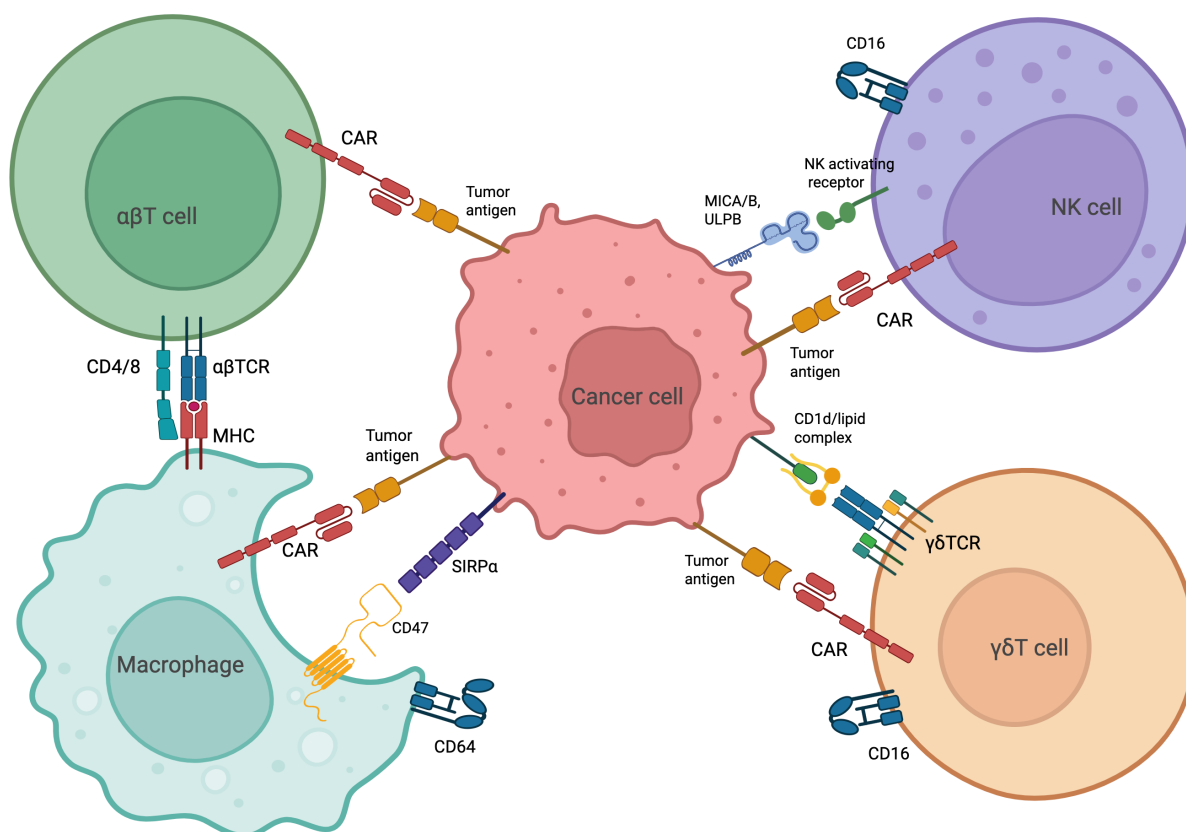


Figure 1 Schematic representation of CAR-engineered immune cells (CAR-T, CAR-NK, CAR- $\gamma\delta$ T, and CAR-M) targeting a cancer cell, highlighting their key biological features and mechanisms of tumor recognition and activation. CAR, chimeric antigen receptor; NK, natural killer.

CAR-CD123-NK cells were tested both in vitro and in vivo in acute myeloid leukemia models and showed remarkable efficacy and a safer profile compared with CAR-CD123-T cells.¹⁶

Additionally, CAR-NK cells have a higher allogeneic potential,¹⁷ and can be gathered from healthy donor blood samples and stem cell sources.

They also overcome antigen immune escape due to their MHC-unrestricted feature. This means they can target and kill tumor cells regardless of whether the target antigen is present on the cell surface. A preclinical study showed that ErbB2-CAR-NK cells directed against HER2+ breast cancer cells bypassed the downregulation of the adhesion molecule intercellular adhesion molecule (ICAM)-1, allowing them to kill cancer cells. This was in contrast to trastuzumab treatment, which required co-stimulation with lymphocyte function-associated antigen (LFA)-1 to activate NK cells for ADCC.¹⁸

Various sources of NK cells have been explored, including umbilical cord blood (UCB), peripheral blood (PB), stem cells, and NK cell lines; no cases of CRS or neurotoxicity have been reported in preclinical and clinical studies.^{19–21}

Limited in vivo persistence remains one of the major hurdles for CAR-NK therapies. NK cells display intrinsically shorter lifespan and reduced proliferative capacity after adoptive transfer, largely due to their dependence on exogenous cytokine support and rapid terminal differentiation. Optimizing the manufacturing, enhancing metabolic fitness, and cytokine independence are required to improve CAR-NK persistence.

Preclinical studies also highlighted that tumor resistance to CAR-NK cells is mainly due to a loss of metabolic fitness, leading to NK cell exhaustion. Engineering CAR-NK cells to secrete inflammatory cytokines could be a strategy to reduce NK cell exhaustion.^{22–26} As an example, the interleukin (IL)-21 armoring can significantly enhance NK expansion, persistence, and antitumor activity.²⁷ Moreover, IL-15-secreting CAR-NK cells targeting CD70 in colorectal and pancreatic cancer models effectively eradicated both tumor cells and CAFs.²⁸ Luo *et al* investigated the effects of ‘superkines’, such as neoleukin-2/15, on anti-mesothelin CAR-NK cells, and superior cytotoxicity has been confirmed due to the activation of the IL-2R β signaling cascade.²⁹

The antitumor activity of CAR-NK cells is also hampered by the immunosuppressive molecules of the TME, such as transforming growth factor β (TGF β), which can impair their killing mechanisms, downregulate activating receptors (aNKRs), and overexpress inhibitory receptors (iNKRs). In some cases, persistent activation of aNKRs without proper co-stimulation by iNKRs can result in NK cell exhaustion.³⁰

Another issue lies in CAR’s engineering method; indeed, NK cells face challenges with viral vectors due to both sensitivity to apoptosis and limited expansion potential.^{31–33} As a result, alternative non-viral delivery methods are emerging as attractive alternatives for CAR

cell manufacturing. Transposon-based systems such as Sleeping Beauty and PiggyBac allow stable genomic integration of CAR constructs while avoiding the complexity and regulatory constraints associated with viral production. These platforms also facilitate multiplex engineering approaches, enabling the simultaneous insertion of CAR constructs together with cytokine-support modules or immune-evasive edits. In parallel, mRNA-based CAR delivery through electroporation or lipid nanoparticles (LNPs) provides transient CAR expression without genomic integration, improving safety and enabling flexible dosing strategies. Although transient expression may require repeated administrations, this approach may be particularly suitable for platforms, such as CAR-NK, which naturally exhibit shorter persistence in vivo. In addition, clustered regularly interspaced short palindromic repeats (CRISPR)-based targeted integration enables precise CAR insertion into defined genomic loci, improving expression stability and reducing the risk of insertional mutagenesis. Collectively, these technologies may substantially accelerate the development of next-generation CAR platforms and facilitate scalable manufacturing for both autologous and off-the-shelf cellular therapies.^{23 34–37}

A further challenge is the phenomenon of trogocytosis, where immune cells, particularly NK cells, exchange surface molecules with antigen-presenting cells (APCs) and tumor cells. This mechanism can lead to tumor cells acquiring inhibitory molecules like programmed death 1 (PD1), which bind to programmed death ligand 1 (PDL1) on NK cells and inhibit their function. Trogocytosis-induced fratricide is another limitation of CAR-NK cell expansion, although KIR-based inhibitory CARs may offer a potential solution to mitigate this issue.³⁸

Lastly, the CAR binding epitope’s location, distant from the NK cell surface, creates magnetic resistance that can hinder antigen binding.³⁹ A recent preclinical study, where CAR-transduced and untransduced NK and T cells targeting CD19 were compared, demonstrated that allogeneic CAR-NK cells exhibited markedly reduced cytotoxicity compared with autologous CAR-T cells against CD19-lymphoblastoid cell lines, whereas untransduced NK cells displayed superior killing ability.⁴⁰

Table 2 summarizes the main challenges for CAR-NK cells.

Advancements in solid tumors

Currently, eight CAR-NK cell clinical trials for solid tumors have been published with results (table 1).

In addition to the block of immunosuppression and the armored CARs, ongoing research is focusing on dual-targeting CAR-NK cells. Zhi *et al* designed CAR-NK cells directed against PDL1 and MICA/B and exploited the block of the PD1/PDL1 axis to achieve a synergistic effect and increase NK cell cytotoxicity, detected by a higher release of perforins and granzymes and a more evident regression of H1299 lung cancer cell lines. They also observed that pyroptosis was the principal way by which

Table 2 Comparison between conventional CAR-T cells and the other CAR-engineered cells

Features	CAR-T $\alpha\beta$	CAR-NK	CAR-T $\gamma\delta$	CAR-M
Cytotoxicity	+++	++	+++	++
Off-the-shelf potential	+	+++	++	+++
Abundance in the body	+++	++	+	++
Adaptability to the antigen	+	+++	+++	+++
Prolonged persistence	++	+	+++	++
Resistance to TME suppression	+	++	+++	+++
Penetrability in the tumor core	+	+	++	+++
Toxicity	+++	+	+	++
Ease of manufacturing	++	+	++	++

low (+), medium (++), high (+++).
CAR, chimeric antigen receptor; TME, tumor microenvironment.

NK cells killed cancer cells, and that their efficiency was higher in the presence of both target antigens, suggesting an AND logic-gated mechanism.⁴¹

In another preclinical study, Jo *et al* improved CAR expression stability on NK cells with entinostat, a histone deacetylase inhibitor, resulting in a more prolonged survival and a higher antitumoral activity.⁴²

Among emerging antigens, mesothelin is overexpressed in breast, ovarian, cervical, and gastric cancers, and treatment with anti-mesothelin CAR-NK cells has shown encouraging results^{43–45}; a phase I clinical trial in China is currently evaluating anti-mesothelin CAR-NK cells against ovarian cancer (ChiCTR2100048100).

B7-H3, overexpressed in solid tumors, is an immunomodulatory transmembrane protein that promotes carcinogenesis and metastasis; basic research is highly focusing on this target with encouraging results.^{46 47}

Several preclinical studies have focused on glioblastoma due to its association with a range of antigens,^{48–50} and a clinical trial with CAR-NK cells targeting HER2 in glioblastoma has shown favorable results (table 1). Strassheimer *et al* found that in untreated HER2-positive glioblastomas, a combinatorial treatment with anti-HER2 CAR-NK cells and anti-PD1 mAbs determines a synergistic antitumor effect, a prolonged survival, and an improved infiltration of T and NKT cells.⁵¹ Furthermore, Gergely *et al* highlighted the importance of CD44, an extracellular matrix TME component, as a critical marker of resistance to trastuzumab treatment in HER2+solid tumors; they demonstrated that anti-HER2 CAR-NK cells have a robust antitumor efficacy against CD44, but not against CD44+HER2 solid tumors.⁵²

Two additional oncofetal glycoproteins, carcinoembryonic antigen (CEA) and glypican-3 (GPC-3), have gained traction in ACT. While CEA is non-specific and has prognostic value in clinical practice, GPC-3 is typically associated with hepatocellular carcinoma (HCC).^{53 54}

Prostate-specific membrane antigen (PSMA), a membrane protein highly expressed in prostate cancer, plays a key role in enzymatic regulation and angiogenesis.⁵⁵

A phase I clinical trial investigating PSMA-directed CAR-NK cells has been conducted in one patient (case report), who achieved SD with prostate specific antigen (PSA) reduction of 97.5% and did not experience any toxicity (NCT03692663).

DLL3, an oncofetal TSA involved in Notch signaling regulation, has also been the focus of CAR-NK research on lung cancer, with a phase I trial currently underway in China (NCT05507593).⁵⁶

A hybrid cell platform that deserves to be discussed is CAR-NKT cells, a rare subset of innate-like immune cells with both T and NK cell receptors. Their superiority over CAR-NK cells is related to the dual antigen targeting function (CD1d-lipid complex and CAR targets), the longer lifespan, and the epitope spreading capacity. In an *in vivo* preclinical study, Dotti *et al* documented a superior antitumor activity of CAR-NKT over CAR-T cells due to the CD1d-dependent immune responses in the TME, eliminating CD1d-expressing M2-like macrophages.⁵⁷ A phase I clinical trial that evaluated GD2-CAR.15 NKTs in 12 heavily pre-treated children with neuroblastoma has been published with results, showing an objective response rate in 25% (3/12) of patients with one complete response and two partial responses (NCT03294954).

CAR- $\gamma\delta$ T CELLS

Since they share a similar repertoire of surface receptors and killing pathways with NK cells, $\gamma\delta$ T cells represent another immune cell subset suitable for CAR engineering, thanks to their hybrid phenotype that bridges innate and adaptive immunity. They can directly recognize stress-induced molecules and non-peptide antigens on tumor cell surfaces, enabling MHC-independent killing. On recognizing tumor cells, $\gamma\delta$ T cells initiate a rapid cytotoxic response by releasing perforin and granzymes, which directly kill target cells. Additionally, they express Fas ligand (FasL) that binds to Fas on tumor cells, triggering apoptosis, and they produce IFN- γ and tumor

necrosis factor (TNF)- α , supporting a coordinated and robust anti-cancer immune response.⁵⁸

Given their localization in mucosal sites and natural tropism for the TME, $\gamma\delta$ T cells exhibit strong infiltration capabilities. After activation, $\gamma\delta$ T cells can acquire phenotypic characteristics of APCs, enabling them to uptake tumor antigens and cross-present them to $\alpha\beta$ T cells, thereby enhancing the antitumor response. This dual functionality allows $\gamma\delta$ T cells to retain their innate antitumor properties while migrating effectively toward tumor sites.⁵⁹

Moreover, $\gamma\delta$ T cells can recognize a unique category of phosphorylated carbohydrate metabolites known as phosphoantigens, which are produced through both endogenous and exogenous pathways under dysregulated conditions, such as chronic inflammation or altered homeostasis. A typical endogenous phosphoantigen is isopentenyl pyrophosphate (IPP), generated by overstimulation of the mevalonate pathway. The intracellular accumulation of IPP can be increased by administering aminobisphosphonates, which inhibit farnesyl pyrophosphate synthase—an enzyme that regulates IPP levels in the mevalonate pathway.⁶⁰

Since there is no standardized expansion protocol, CAR- $\gamma\delta$ T cells are manufactured similarly to conventional CAR-T and CAR-NK cells. Like NK cells, $\gamma\delta$ T cells can be sourced from PB, UCB, and immortalized cell lines. For activation and expansion, phosphoantigens or aminobisphosphonates, such as zoledronic acid, may be used, in addition to inflammatory cytokines.

The V γ 9V δ 2 subset is the predominant $\gamma\delta$ T subgroup in PB and can be selectively expanded in vitro and in vivo using T cell mitogens (eg, concanavalin A) or aminobisphosphonates. V γ 9V δ 2 lymphocytes target butyrophilin (BTN) proteins on tumor cell surfaces (specifically BTN3A1/2A1) and exhibit an enhanced ability to acquire an APC phenotype. The second most prevalent subgroup, V γ xV δ 1 (V δ 1) lymphocytes, binds to CD1 molecules on tumor cells and can be expanded using T cell mitogens, artificial APCs, and anti-CD3 antibodies. This subgroup is distinguished by a naïve memory phenotype, a high prevalence in mucosal tissues, lower susceptibility to inhibitory receptors and cell death mechanisms, and less exhaustion (especially with concanavalin A). Protocols for expanding $\gamma\delta$ T cells require pro-inflammatory cytokines (IL-2, IL-7, IL-15) to sustain their proliferation and functionality.^{60 61} In contrast, Bridge *et al* have developed a feeder-free, antibody-based, ex vivo pan-expansion protocol for $\gamma\delta$ T cells, with a transposon-based CAR integration and the inclusion of soluble IL-15. With this method, $\gamma\delta$ T cells were efficiently expanded and activated in vivo and have been detected in almost all tissue samples, including the brain, indicating a great ability of penetration and tumor trafficking.⁶²

Advantages and drawbacks

CAR- $\gamma\delta$ T cells represent an advancement over CAR-NK cells, as they not only share NK cells' advantages but also

function as APCs and can recognize phosphoantigens, even if their lower abundance in the body and challenges in expansion remain significant limitations.

Wang *et al* demonstrated that V δ 2 T cells, expanded from healthy donor PBMCs using a novel bisphosphonate prodrug (PTA) combined with IL-7 and IL-15, and subsequently transduced with a CEA-specific CAR, effectively suppressed tumor growth in a xenograft model of CEA-expressing solid tumors. Their antitumor activity was further enhanced by introducing the GTR ligand, which helped prevent exhaustion.⁶³ While the V δ 2 subset has been extensively studied, the V δ 1 subset has remained challenging due to the absence of a standardized expansion protocol. A method adapted by Ferry *et al* for V δ 1 T lymphocytes, based on the CAR- $\alpha\beta$ T cell protocol, utilized anti-CD3 OKT-3 stimulation of PBMCs with IL-15, achieving high expansion rates and rendering V δ 1 T cells amenable to CAR transduction.⁶⁴

Concerning non-viral delivery methods, studies have shown that electroporating NKG2Dz RNA CAR can enhance the antitumor activity of V γ 9V δ 2 T cells in both in vitro and in vivo models of solid tumors. In one study, this approach has delayed more tumor progression than unmodified V γ 9V δ 2 T cells expanded with Zometa, extending the median survival time of treated mice by up to 132% compared with the PBS control group.^{65 66}

The major disadvantage of CAR- $\gamma\delta$ T cells is the low abundance in the body, but the prolonged persistence and the resistance to TME suppression make them a promising cell platform for solid tumor treatment (table 2).

Advancements in solid tumors

Currently, neither in hematologic nor in solid tumors clinical trials with CAR- $\gamma\delta$ T cells have been published. Early clinical signals from IL-2 receptor-engineered $\gamma\delta$ T-cell approaches in pediatric acute myeloid leukemia highlight the feasibility of engineering innate-like lymphocytes with improved persistence and proliferative capacity.⁶⁷ Five clinical trials are ongoing, with two of these testing anti-CD20 CAR- $\gamma\delta$ T cells directed against various types of B-cell malignancies (NCT04911478 and NCT04735471), another one testing HLA-G CAR.BiTE allogeneic $\gamma\delta$ T cells in advanced solid tumors (NCT06150885), and the other two testing allogeneic anti-NKG2DL CAR- $\gamma\delta$ T cells against advanced solid tumors (NCT05302037 and NCT04107142).

One of the earliest preclinical studies on CAR- $\gamma\delta$ T cells for solid cancers focused on melanoma. In this study, $\gamma\delta$ T cells, expanded with zoledronate or OKT3 and engineered with either an mRNA-electroporated gp100/HLA-A2-specific TCR or a CAR targeting MCSP, demonstrated robust in vitro cytotoxicity, along with reduced cytokine production.⁶⁸ Makkouk *et al* developed CAR-V δ 1 T cells directed against GPC-3 and co-expressing soluble IL-15, expanded from allogeneic PBMCs. These GPC-3 CAR-V δ 1 T cells exhibited a naïve-like phenotype and demonstrated potent in vitro and in vivo cytotoxicity against HCC cell lines and xenograft models. They

showed favorable safety profiles and limited expression of exhaustion markers, suggesting that the V δ 1 T cell subset might offer superior antitumor activity both qualitatively and quantitatively.⁶⁹ Claudin 18.2 (CLDN18.2), a TAA overexpressed in gastric cancer, already studied in a clinical trial with conventional CAR-T cells, has been the focus of a recent preclinical study, where CAR- $\gamma\delta$ T cells exhibited superior cytotoxicity with a prolonged survival of mice, compared with traditional CAR-T cells.⁷⁰ In another preclinical study, CD16^{hi} V δ 2 T cells, engineered with a mesothelin-targeting CAR and IL-15, showed impressive levels of cytotoxicity through CAR, TCR, and ADCC-mediated mechanisms. These modified cells demonstrated sustained persistence and high safety in two ovarian cancer models, without inducing GvHD.⁷¹ Additionally, genetically engineered V γ 9V δ 2 T cells with CAR-B7H3 demonstrated promising efficacy across various glioblastoma specimens.⁷²

CAR-MACROPHAGES

Macrophages are among the most abundant immune cells in the TME. Depending on the signals, macrophages can adopt an anti-tumor (M1) or a tumor-promoting (M2) phenotype.⁷³ M1 macrophages exhibit tumoricidal activity by releasing pro-inflammatory cytokines and upregulating markers involved in antigen presentation through MHC II. In contrast, M2 macrophages have an immunoregulatory role, releasing anti-inflammatory cytokines and promoting tumor growth and metastasis. Previous studies using adoptive transfer of unmodified macrophages were unsuccessful, as these cells often differentiated into the M2 phenotype.⁷⁴ While CAR-Ms share some challenges with CAR-T cells, such as the need for specific antigen targeting and the risks of antigen escape and systemic cytokine toxicity, they have unique advantages in overcoming two major hurdles in solid tumor treatment: immune cell trafficking and infiltration into the TME, as well as resistance to the immunosuppressive TME.

Advantages and drawbacks

Klichinsky and colleagues⁷⁵ transduced the human macrophage Tohoku hospital pediatrics (THP)-1 cell line with a first-generation anti-CD19 CAR encoding the CD3 ζ signaling domain; this CAR19 ζ macrophage demonstrated antigen-specific phagocytosis of CD19-expressing tumor cells. Similar results were achieved with CAR-Ms targeting solid tumor antigens like mesothelin and HER2. In another study, Zhang *et al.*⁷⁶ examined various signaling domains to adapt CAR function to macrophages' innate pathways. CAR-Ms with CD3 ζ or Fc γ R signaling domains exhibited similar efficacy, likely due to CD3 ζ 's structural similarity to Fc ϵ RI- γ , a key signaling molecule in macrophage antibody-dependent phagocytosis. However, using DAP12 as a signaling domain reduced phagocytic activity compared with conventional signaling domains,⁷⁷ potentially due to differences in the immunoreceptor

tyrosine-based activation motifs between CD3 ζ and DAP12. One of the primary challenges in manufacturing CAR-Ms is their inherent ability to recognize and react to foreign nucleic acids, making them resistant to genetic manipulation.⁷⁸ This characteristic complicates the use of lentiviral transduction methods when applied to macrophages. To address this issue, Bobadilla *et al.*⁷⁹ introduced a workaround using lentiviral particles derived from HIV-1 with the viral accessory protein Vpx, which facilitates the degradation of the myeloid-specific HIV-1 restriction factor. An alternative approach has been adenoviral transduction using a chimeric adenovirus 5-fiber 35 vector (Ad5f35), which effectively delivers genes to macrophages by leveraging their CD46 expression.⁸⁰ This method not only achieves successful gene insertion but also activates the inflammasome, locking macrophages into an M1-like phenotype.⁷⁵ More recently, non-viral methods, such as intratumoral injection of nanoparticles loaded with plasmid DNA encoding an anti-HER2 CAR, successfully transfected macrophages within a patient-derived xenograft model.⁸¹ In another study, Ye *et al.*⁸² demonstrated that CAR-Ms delivered via this system achieved significant anti-tumor cytotoxicity. However, information about the persistence of CAR expression in macrophages after delivery remains limited. Studies suggest that mRNA-transfected T cells typically express CARs for about a week.⁸³ To address this, researchers are exploring repetitive mRNA delivery using nanobodies, aiming to maintain prolonged CAR expression in host lymphocytes, potentially achieving comparable *in vivo* efficacy to retrovirally transduced CAR-T cells.

Several studies highlight different advantages that CAR-Ms offer over CAR-T and CAR-NK cells. Like CAR- $\gamma\delta$ T cells, CAR-Ms can effectively migrate to the tumor site and infiltrate the TME. Furthermore, they can broaden the immune response by recruiting other immune cells and reprogramming the TME through polarization from M2 to M1 phenotype. Similarly to NK and $\gamma\delta$ T cells, CAR-Ms can be generated from different sources, including PB, UCB, hematopoietic stem and progenitor cells, induced pluripotent stem cells (iPSCs), and THP-1 cells, which offer flexibility in their manufacturing. Although clinical trial data is currently limited, CAR-Ms may demonstrate a safer profile *in vivo* due to their shorter circulation time. Despite their potential, CAR-Ms have limitations, particularly their low proliferation rate.⁸⁴ They do not naturally expand *in vivo*, requiring repeated infusions to maintain sufficient cell numbers for effective immunosurveillance. Another important challenge is their 'sticky' behavior, due to the expression of a wide amount of integrins and other adhesion molecules. In particular, the latter one could result in suboptimal distribution if CAR-M accumulate in specific areas without sufficient dispersion.

Advancements in solid tumors

Most CAR-Ms studies are still in the preclinical phase, but they have already shown promising results in both hematological and solid tumors (table 1).

CAR-Ms-mediated cytotoxicity primarily occurs through cancer cells' phagocytosis. Morrissey *et al.*⁸⁵ showed that CD19-targeted CAR-Ms successfully eliminated CD19+beads as well as Raji B cells in vitro. Moreover, CD47-mediated signaling inhibition caused tumor regression in ovarian and melanoma cancer models both in vitro and in vivo.^{86 87} In another study, CD147-CAR-Ms engineering demonstrated enhanced M1 polarization and phagocytic activity while maintaining a safe profile, showing minimal phagocytosis of normal cells despite their expression of the CD147 antigen.⁸⁸ Targeting c-MET is another strategy adopted in pancreatic cancer, where CAR-Ms have demonstrated to disrupt angiogenesis and to engulf both MET+cancer cells and cancer stem cells.⁸⁹

Adenovirally transduced anti-HER2 CAR-Ms also exhibited antigen-specific phagocytosis and dose- and time-dependent killing of HER2+tumor cells, while cross-presenting intracellular tumor antigens to T-helper cells.⁷⁵ Additionally, in NSG mice, anti-HER2 CAR-Ms persisted for up to 62 days, resulting in prolonged overall survival. CAR-Ms demonstrated epitope spreading by cross-presenting intracellular tumor antigens to T cells, which enhanced their anti-tumor response. For instance, anti-HER2 CAR-M treatment in CT26-bearing mice increased intratumoral CD4+ and CD8+ T cell populations and improved T cell reactivity to the CT26 antigen gp70.⁹⁰ Beyond these functions, CAR-Ms also recruit dendritic cells into the TME, while repressing Tregs, M2 macrophages, and MDSCs.⁸¹ In this context, the first-in-human Phase I clinical study of the HER2-targeted CAR-macrophage (CT-0508) represents a key milestone for the field, demonstrating clinical safety and early evidence of antitumor activity in patients with advanced HER2-overexpressing solid tumors.⁹¹ Importantly, no lympho-depletion chemotherapy was used before infusion.

An innovative approach to generate CAR-Ms is by deriving human iPSCs, providing a potentially unlimited source of engineered immune cells. This approach was applied in a pancreatic cancer model targeting prostate stem cell antigen (PSCA), where CAR-Ms showed strong antitumor activity and improved survival in mouse models.⁹²

Another innovative approach has been the inhibition of furin, a proprotein convertase known to be overexpressed in solid tumors and to reprogram macrophages to an M2 phenotype. Furin-deficient CAR-Ms demonstrated superior phagocytic activity and efficient immune cell recruitment.⁹³

Finally, LNP technology has been recently adapted in HCC models, where CAR-Ms generated via LNPs, combined with CD24-Siglec-G blockade, showed an enhanced phagocytic activity in liver macrophages.⁹⁴

CONCLUSION

Conventional CAR-T therapies have transformed hematology, yet their limited efficacy in solid tumors highlights a biological mismatch. Antigen heterogeneity,

impaired trafficking, and TME immunosuppression collectively restrict CAR-T persistence and functionality. Importantly, the clinical reality of patients with solid tumors (often older, frailer, and less tolerant of lympho-depleting regimens) further narrows the therapeutic window. In this context, CAR-engineered innate and innate-like immune cells may represent more than valid alternatives. CAR-NK and CAR-NKT platforms offer scalable off-the-shelf potential with reduced toxicity, while CAR- $\gamma\delta$ T cells combine tumor tropism with antigen-presenting capacity. CAR-Ms uniquely address one of the most defining barriers in solid tumors: physical and immunologic exclusion from the tumor core, with early clinical studies demonstrating feasibility and safety. Ultimately, the next breakthrough in ACT for solid tumors will likely emerge from rational integration of these platforms with improved manufacturing, non-viral engineering, and combinatorial strategies targeting both cancer cells and TME. Among next-generation platforms, CAR-NKT and CAR- $\gamma\delta$ T cells may be particularly well suited for clinical translation in solid tumors, as they combine innate-like tumor tropism with adaptive-like functionality, potentially enabling improved persistence, antigen spreading, and resistance to TME suppression. While clinical validation remains early, these hybrid cellular populations could represent the most impactful evolution beyond ACT.

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